★ WEEK 10 · LECTURE

How Not to Fool Yourself

Backtesting pitfalls, purged CV — and the Deflated Sharpe

Expanded edition · 38 slides · with three hands-on labs

The most dangerous person in finance is the one fooled by their own backtest.
This week is about honesty: testing strategies without secretly cheating.

Where We Are Going Today

Week 1 said process beats genius. This is the process that keeps you honest

1

Why backtests lie

A backtest is a practice exam you can secretly peek at. First, how we even score a strategy.

2

The time-machine problem (leakage)

Shuffling time data lets the model see the future. Fix: purged cross-validation.

3

Fooled by trying too much

Test enough ideas and one wins by luck. Fix: the Deflated Sharpe Ratio.

By the end you will purge a timeline by hand and compute luck's par score for 1,000 tries.

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PART 1 · THE SELF-DECEPTION

The Great Self-Deception

Why backtests lie

A Backtest Is a Practice Exam

...one where it's frighteningly easy to peek at the answers

A backtest replays a strategy on past data to see how it 'would have' done. The problem: with a few careless choices, you accidentally let it see answers it never could have had.

Why it's so seductive

- A beautiful equity curve feels like proof.
- It's easy to keep tweaking until it looks great.
- Almost every published strategy looks amazing on paper.

The honest mindset

- Assume a great result is a mistake until proven otherwise.
- The goal is not a pretty curve — it's a curve you can TRUST.
- Today: the two biggest ways the curve lies.

Backtests Don't Predict the Future

López de Prado: most backtested strategies are false discoveries

"Most quant strategies that look brilliant in a backtest are statistical flukes.
A backtest is not an experiment — it's a simulation you can rig."

What goes wrong

- Leakage: the model peeks at future data.
- Over-tweaking: tuned until it fits the past perfectly.
- Multiple testing: the lucky winner of many tries — all three inflate the backtest, then vanish live.

This week's two fixes

- Purged cross-validation → kills leakage.
- Deflated Sharpe Ratio → discounts luck from many tries.
- Together: a backtest you can actually believe.

What Pure Luck Can Manufacture

Four numbers from this week's own Lab 2 — zero skill anywhere

1,000

Coin-flip strategies our Lab 2 generates at random

~50

Expected to clear a naive 5% 'significance' bar by chance

≈ 1.9

Annualized Sharpe of the BEST pure-luck strategy (3 years)

0

Actual skill in every single one of them

Hold that 1.9: any backtest Sharpe must first beat what luck alone would deliver.

First, How We Score a Strategy: Sharpe

Reward per unit of risk — the standard yardstick

$$\text{Sharpe} = \frac{\text{average return}}{\text{volatility (risk)}}$$

Reading it

- Top: how much the strategy earns on average.
- Bottom: how bumpy the ride is (its volatility).
- A high Sharpe = good return without wild swings.

Why it can mislead

- A backtest Sharpe of 3 looks incredible…
- …but it may be leakage or luck, not skill.
- The rest of today asks: is this Sharpe REAL? We must deflate it.

A Sharpe Ratio by Hand

Five monthly returns — mean, wobble, and the yardstick (full version: problem set)

Months	+2%, −1%, +3%, 0%, +1%	Deviations from mean 1%: +1, −2, +2, −1, 0
Volatility	σ = √[(1+4+4+1+0)/5] = √2 ≈ 1.41%	the wobble around the average
Sharpe (monthly)	1 / 1.41 = 0.71	reward per unit of wobble
Sharpe (annualized)	0.71 × √12 ≈ 2.45	the number people quote

2.45 looks superb — and today's whole question is whether a number like that survives an honesty audit.

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PART 2 · LEAKAGE

The Time-Machine Problem

Leakage and purged cross-validation

Shuffling Time = Seeing the Future

Ordinary cross-validation shuffles data — but markets are a timeline. Training after testing = tomorrow's newspaper.

Shuffling Time Data = Reading Tomorrow's Newspaper

Wrong: shuffle the days



Train (green) and test (orange) days are all jumbled together.

The model trains on days that come AFTER the test days.

It peeks at the future. Fake great score.

Right: keep time in order



Train on the PAST (green), then test on the FUTURE (orange).

The model never sees data from after the moment it predicts.

Honest score — just like real trading.

Why Standard Cross-Validation Fails Here

The trick that works everywhere else breaks on time data

Normal k-fold CV randomly splits data into folds — fine for photos of cats. But for a timeline, a random split mixes past and future, leaking information you couldn't really have.

Two sources of leak

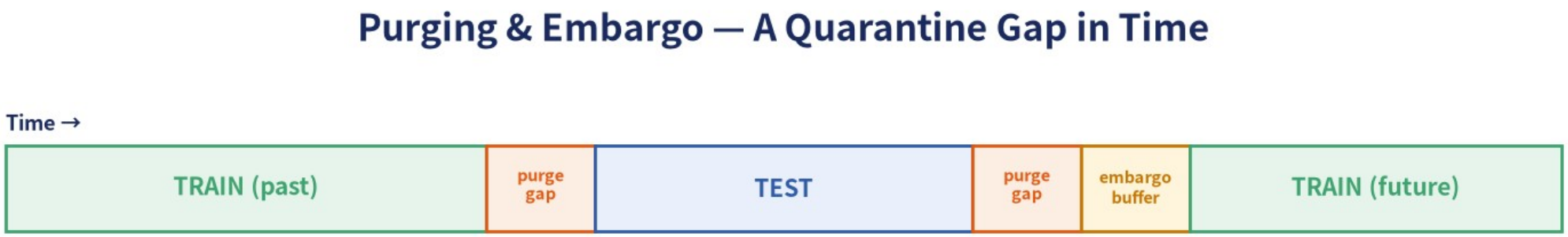
- Order: training on days after the test day.
- Overlap: a label spanning the next 10 days touches both sides.
- Both make the score look better than reality.

The cure

- Always split by TIME (train past, test future).
- Then PURGE the overlapping training samples.
- Add an EMBARGO buffer after the test set.

Purging & Embargo

Quarantine the days that touch both sides of the train/test line. Purge the overlap; embargo a buffer.



Purge: remove the overlap on both sides

A training label that touches the test labels' span leaks information, before or after the test. Drop every such training day.

Like a quarantine: anyone touching the span is out.

Embargo: a cooling-off buffer

Prices stay correlated for a while after the test labels end. So also skip a short stretch right after that span.

Together they give an honest, leak-free test.

Purging a Timeline by Hand

Ten days, 3-day label windows, test = days 5–7, embargo = 1 day (full version: problem set)

Day	1	2	3	4	5	6	7	8	9	10
Label window	1–3	2–4	3–5	4–6	5–7	6–8	7–9	8–10	9–11	10–12
Verdict	train	train	PURGE	PURGE	test	test	test	PURGE	PURGE	EMBARGO

Test labels reach day 9, so days 3–4 and 8–9 are purged; day 10 sits out the embargo. Honest training set: {1, 2} — two days, not seven.

Purged Cross-Validation in Practice

The standard, leak-free way to test a market strategy

The steps

- Split the timeline into ordered chunks; train on some, test on a later one.
- Purge training samples whose labels overlap the test labels' span, on both sides.
- Embargo a short stretch after the test labels end.

What you gain

- A score that reflects real, tradeable performance — no peeking.
- Usually a LOWER score — but an honest one.
- A lower honest number beats a high fake one, every time.

scikit-learn's TimeSeriesSplit enforces time order; purging adds the overlap fix on top.

Three Splitters, One Honesty Ladder

What each tool protects you from — and what it still misses

Protection needed	Shuffled KFold	TimeSeriesSplit	Purged CV + embargo
Train stays in the past	× — mixes past & future	✓	✓
Overlapping labels removed	×	× — boundary still leaks	✓ (purge)
Buffer after the test set	×	×	✓ (embargo)
Verdict for market data	dangerous	better — still leaky at edges	the honest standard

Our Lab 1 measures rung 1 vs rung 2; Lab 3 has Claude build rung 3 in full.

The Leakage Rogues' Gallery

Six ways the future sneaks into training — most look completely innocent

The leak	Innocent disguise	The fix
Shuffled CV on a timeline	'standard practice everywhere else'	split by time, always
Label overlap at the boundary	'my split IS chronological'	purge + embargo (today)
Normalizing with full-sample stats	'I just standardized the data'	fit scalers on train only
Survivorship bias (Week 2)	'I used today's index members'	point-in-time universe
After-the-fact features	'this fraud flag is so predictive!'	decision-time info only
Re-tuning after seeing the test	'one last tweak'	freeze the test set — once

One question catches them all: could I really have known this, at that moment?

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PART 3 · MULTIPLE TESTING

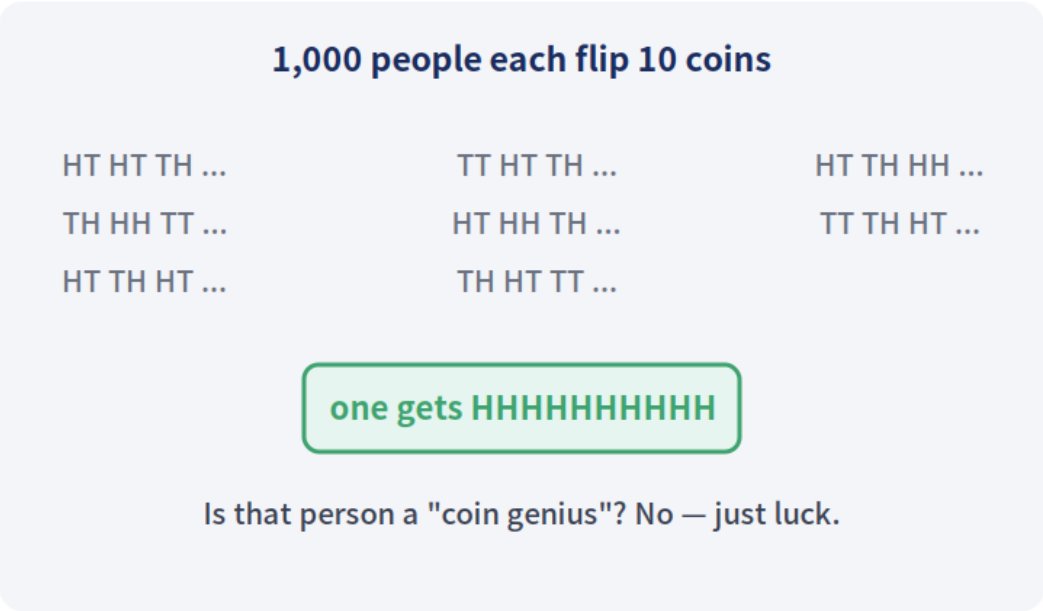
Fooled by Trying Too Much

Multiple testing & the Deflated Sharpe

The Coin-Flip Trap

Try enough strategies and one looks brilliant by pure luck. The factor zoo (Week 4): 1,000 ideas, ~50 dazzle by chance.

Try Enough Strategies and One Wins by Pure Luck



The same trap in finance

Test 1,000 strategy ideas on past data.

A few will look spectacular — by luck alone.

You proudly publish the lucky winner...

...then it loses money the moment it's live.

The more you try, the higher the bar should be before you believe a result.

The More You Try, the Higher Luck Climbs

A formula for how good the best fluke looks

$$\mathbb{E}[\max \text{ Sharpe by luck}] \approx \frac{1}{\sqrt{T}}\left[(1-\gamma)\,\Phi^{-1}\left(1-\tfrac{1}{N}\right)+\gamma\,\Phi^{-1}\left(1-\tfrac{1}{Ne}\right)\right]$$

Reading it

- N = strategies, T = data points, γ ≈ 0.58.
- Even with ZERO real skill, the best one looks this good by luck.
- It grows with N — more tries, higher fake Sharpe.

The lesson

- A Sharpe of 2 after 1 try is impressive.
- A Sharpe of 2 after 1,000 tries may be nothing.
- Judge a result against how hard you searched — exactly what the DSR does.

Luck's Par Score, Computed by Hand

Expected best of N on three years of daily data, annualized (full version: problem set)

Trials N	Bracket (noise units)	Daily luck Sharpe (÷√756)	Annualized (×√252)
10	1.58	0.057	≈ 0.91
100	2.53	0.092	≈ 1.46
1,000	3.25	0.118	≈ 1.88

Luck's par rises with every try: after 1,000 experiments, a Sharpe of 2 barely clears what NOTHING looks like.

The Deflated Sharpe Ratio (DSR)

Given how many strategies I tried, does this Sharpe beat what luck alone delivers?

$$\text{Deflated Sharpe} = \Phi\left(\frac{\text{SR} - \text{SR}^*}{\text{standard error}}\right)$$

How it works

- Take the raw backtest Sharpe.
- Subtract the 'luck threshold' that grows with the number of trials.
- Convert what's left into confidence against luck.

How to read it

- DSR above 0.95: beats luck — worth pursuing.
- DSR near 0.5: what luck alone delivers — walk away.
- It punishes over-searching; an honest quant always reports the trial count.

One Sharpe, Three Verdicts

The same backtest Sharpe of 2.0 — judged against how hard we searched

How many tries?	Luck's par score	Sharpe 2.0 vs par	Verdict
N = 10	0.91	well above	DSR 0.97 — beats luck at 95%
N = 100	1.46	above	DSR 0.82 — promising, not proven
N = 1,000	1.88	barely above	DSR 0.58 — no better than luck

The number never changed — only the honesty about the search did. This is the DSR's whole worldview.

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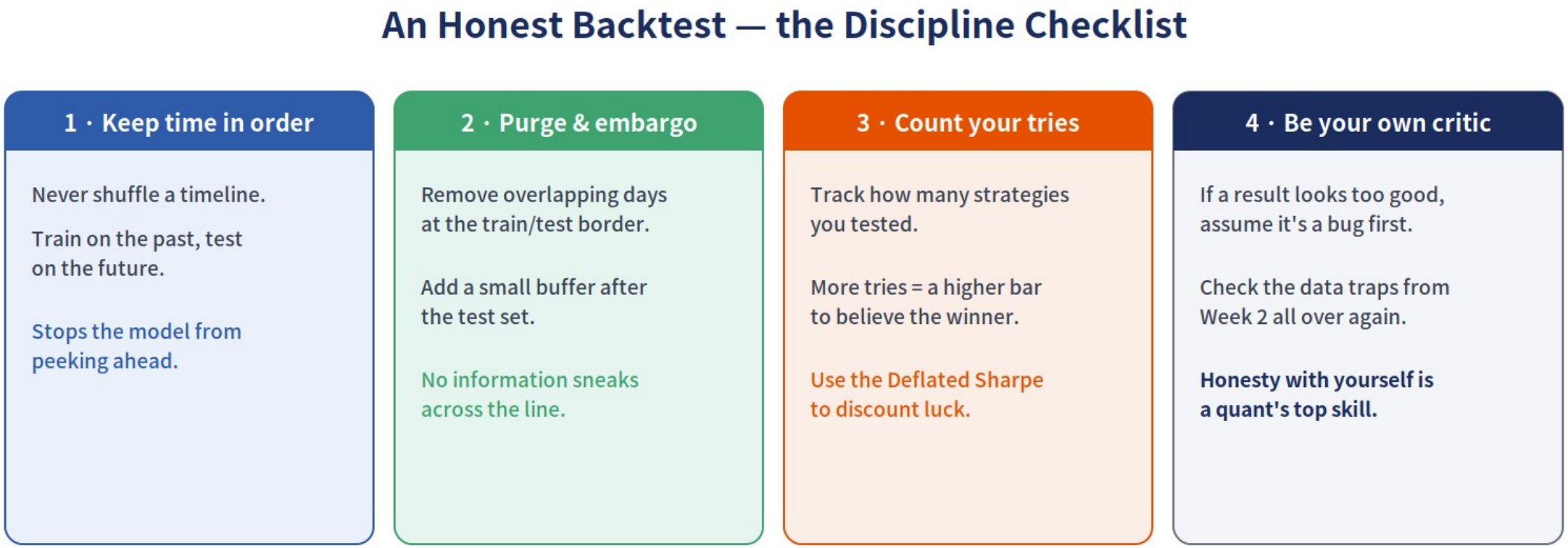
PART 4 · DOING IT RIGHT

Doing It Right

The discipline that keeps you honest

The Honest-Backtest Checklist

Four habits that separate real strategies from flukes. None are fancy — and discipline is what survives live.



The Honest Quant's Working Rhythm

Three habits that make the checklist automatic

1 · Log every trial

before you run it

Every variant you test goes in the ledger — including the embarrassing ones. N is a fact, not a feeling.

the DSR needs the TRUE N

2 · Freeze the test set

touch it once

Tune on train/validation only. The test window is opened one time, at the very end — like a real exam.

re-tuning after peeking = leak #6

3 · Report the deflated number

Sharpe + N + T together

Quote the Sharpe, the trial count, the record length AND the DSR. A naked Sharpe is an advertisement, not evidence.

'SR 2.0, N = 400, 3 y, DSR 0.68'

Same spirit as Week 1: a process you can defend beats a genius story you can't.

The Same Trap in Our Three Apps

Leakage and luck hide everywhere — not just in trading

Fraud detection

the after-the-fact leak

A feature only known AFTER fraud is confirmed makes the model look perfect — then it fails live.

fix: decision-time info only

Recommendation

the A/B luck trap

Try 500 tweaks, ship the best A/B result — it may be noise from many tries.

fix: count experiments, raise the bar

Trading

both at once

Labels bleed across the split AND the signal is the best of thousands.

fix: purged CV + DSR, every time

The discipline is universal: only use what you'd really know, and never confuse luck with skill.

The Honesty Thread Through the Whole Course

Week 10 is not a new topic — it is the course's spine, made explicit

Week	The self-deception	The antidote we learned
W1	memorizing noise looks like genius	train/test split, the U-curve
W2	dead companies vanish from history	survivorship-aware data (12.4pp gap!)
W4	the factor zoo dazzles	regularization; suspicion of many factors
W9	random correlations look like structure	the Marchenko-Pastur noise edge
W10	leakage and lucky winners	purged CV + the Deflated Sharpe

One enemy all along: mistaking what randomness can produce for what skill did produce.

Spot the Pitfall — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

Scenario	Pitfall & fix
Shuffled 5-fold CV scores 0.9; live trading flops	Leakage by order — split by time instead
Chronological split, but 10-day labels at the edge	Boundary leak — purge + embargo
Best of 800 signals over 3 years has Sharpe 1.9	Luck's par ≈ 1.8 — barely above, deflate first
Scaler fit on ALL data before splitting	Subtle leak — fit preprocessing on train only
'One last tweak' after seeing test results	Frozen test violated — the exam is over

If you got 4 of 5, the auditor's instinct of Week 10 is yours.

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PART 5 · HANDS ON

Your Turn — Hands On

Watch leakage and luck inflate scores with your own eyes

Getting This Week's Data

Companion notebook — the price/return series (§2)

From the companion notebook (§2)

- Run Setup (§0), then the §2 loader.
- Gives px (daily prices) and ret (daily returns).
- Same series we first met in Week 1; offline backup in §5.

What we'll do

- Lab 1: show leakage — shuffled CV vs time-ordered CV.
- Lab 2: multiple testing — luck inflates the best Sharpe.
- Both in numpy + scikit-learn, a few lines each.

[CODE] FML_Data_Setup_Guide.ipynb · §2 price/return loader

```
import yfinance as yf
px  = yf.download('AAPL', period='3y', auto_adjust=True,
                  multi_level_index=False, progress=False)['Close']
ret = px.pct_change().dropna()      # daily returns
```

Hands-On Lab 1 — Watch Leakage Inflate a Score

Shuffled CV vs. time-ordered CV on the same task

[CODE] colab_lab1.py · data from setup guide §2

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import (KFold, TimeSeriesSplit,
                                     cross_val_score)

r = ret.values
X = np.column_stack([r[:-1], np.roll(r, 1)[:-1]])  # past
y = r[1:]                                          # next

shuf = cross_val_score(LinearRegression(), X, y,
         cv=KFold(5, shuffle=True, random_state=0))
time = cross_val_score(LinearRegression(), X, y,
         cv=TimeSeriesSplit(5))
print('shuffled CV (leaky):', round(shuf.mean(), 4))
print('time-ordered CV    :', round(time.mean(), 4))
```

Your task (step by step)

- Build features from past returns.
- Score a model with SHUFFLED k-fold.
- Score it again with time-ordered split.
- Compare the two scores.
- Q: which one is the honest (lower) number?

Hands-On Lab 2 — Luck from Many Tries

Generate random strategies; watch the best one shine

[CODE] colab_lab2.py · data from setup guide §2

```
import numpy as np
from scipy.stats import norm
r = ret.values; n = len(r)
rng = np.random.default_rng(0)

sharpes = []
for _ in range(1000):               # 1000 random strategies
    signal = rng.choice([-1, 1], size=n)  # random buy/sell
    pnl = signal * r
    sharpes.append(pnl.mean() / pnl.std())

best = max(sharpes)
N = 1000; q1 = norm.ppf(1 - 1/N); q2 = norm.ppf(1 - 1/(N*np.e))
luck = (0.42*q1 + 0.58*q2) / np.sqrt(n)  # luck's par
print('best random Sharpe:', round(best, 3))
print('explained by luck :', round(luck, 3))
```

Your task (step by step)

- Make 1,000 RANDOM trading signals.
- Compute each one's Sharpe on real returns.
- Take the BEST Sharpe of all 1,000.
- Compare it to the luck threshold.
- Q: is the 'best' strategy actually skillful?

Hands-On Lab 3 — Claude as Your Coding Copilot

AI is allowed in this course — it explains and extends; you understand and verify

[INPUT] Prompt · Claude (paste after your Lab 1 + 2 code)

```
You are my machine-learning tutor.

1) Explain my Lab 1 and Lab 2 code above, one
   plain-English sentence per line.
2) Implement purged time-series CV: 5 ordered folds,
   3-day label windows, purge overlapping training
   rows on both sides, 5-day embargo; rescore Lab 1's
   model and add a third row (shuffled / time /
   purged).
3) Sweep the trial count: for N in [10, 100, 1000]
   plot best-of-N random Sharpe vs the
   López de Prado expected-max luck curve, annualized.
4) Compute a simple deflated verdict for Lab 2's best
   strategy: is it above or below luck's par?
5) In 3 sentences: why the purged score is the one
   you would show an investor.
```

House rules for AI use

- Allowed: explain, debug, extend code you wrote.
- Required: run it, read it, defend every line.
- Forbidden: submitting code you cannot explain.
- Tip: ask 'why', not just 'do'.

Comprehension Check

Try each question first — the answer is revealed below

Q1 Why is shuffled k-fold CV dangerous for market data?

- A It is too slow
- **B It lets the model train on future data (leakage) ✓**
- C It deletes features
- D It needs labels

Answer: Shuffling mixes past and future, so the model peeks ahead — inflating the score. Split by time instead.

Q2 What does the Deflated Sharpe Ratio correct for?

- A Slow training
- **B Luck from testing many strategies ✓**
- C Missing data
- D Too few features

Answer: DSR discounts the Sharpe for how many strategies you tried — the more tries, the higher the luck bar.

Comprehension Check — Round Two

Two more — these cover the new slides

Q3 Purging and embargo differ how?

- A They are the same thing
- B **Purge removes label-overlap rows; embargo skips a buffer AFTER the test** ✓
- C Embargo removes features
- D Purge shuffles folds

Answer: Purge kills rows whose label windows touch the test labels' span; embargo rests the days right after it. Both guard the border.

Q4 Two teams report Sharpe 2.0. Team A ran 10 trials, Team B ran 1,000. Who is more credible?

- A Team B — more thorough
- B **Team A — luck's par was only 0.91 for them** ✓
- C Equally credible
- D Neither, Sharpe is useless

Answer: Same Sharpe, different search intensity: B's 2.0 barely clears luck's 1,000-trial par of 1.88; A's clears 0.91 easily.

Week 10 on One Page

Photograph this slide — it is the whole lecture

Concept	One line to remember
Backtest	a practice exam you can secretly peek at — assume rigged until audited.
Sharpe	mean/std of returns; annualize by √12 (monthly) or √252 (daily).
Leakage	any future info in training: order, overlap, scalers, survivorship, hindsight features.
Purged CV	split by time → purge label overlaps on both sides → embargo a buffer after the test labels.
Luck's par	the expected best-of-N Sharpe that ZERO skill earns; rises with N, falls with T.
DSR	your Sharpe's confidence against luck, after charging it for the search.

Next stop: building portfolios — with the clean covariance Week 9 gave us.

Three Ideas, One Picture Each

The theme from Week 1, made concrete: process and honesty beat a brilliant-looking backtest

1

Backtests are easy to rig

A pretty equity curve is not proof. Assume a great result is a mistake until you've ruled out cheating.

2

Leakage = seeing the future

Never shuffle a timeline. Split by time, then purge overlaps and embargo a buffer (purged CV).

3

More tries = more luck

Test enough ideas and one dazzles by chance. The Deflated Sharpe discounts your score for trials.

Keep the pictures: a rigged practice exam, a time machine to dismantle — and luck's rising par.

Week 10 — Three Things to Remember

Before next class: finish Labs 1–3 and note the gap between the leaky and honest scores

1

Don't trust a pretty backtest

Most great-looking strategies are flukes. The goal is a result you can TRUST, not one that just looks good.

2

Purged CV stops leakage

Split by time; purge overlapping samples; embargo a buffer. No peeking at the future = an honest score.

3

Deflated Sharpe stops luck

The best of many random tries looks great by chance. Discount your Sharpe for how hard you searched.

Next week — Portfolio Construction: HRP and NCO, using the clean covariance from Week 9.